

lab03(SSU)

September 10, 2025

1 Lab 3: Table Operations

Welcome to Lab 3! This week, we'll learn how to import a module and practice table operations. The [Python Reference](#) has information that will be useful for this lab.

Recommended Reading: * [Introduction to Tables](#)

Submission: Once you're finished, run all cells besides the last one, select File > Save Notebook, and then execute the final cell.

Let's begin by setting up the tests and imports by running the cell below.

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
      !pip install py pandoc
      !pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

The following additional packages will be installed:

```
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronex2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc-data poppler-data preview-latex-style
rake ruby ruby-net-telnet ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0
rubygems-integration tlutils teckit tex-common tex-gyre texlive-base
texlive-binaries texlive-fonts-recommended texlive-latex-base
texlive-latex-recommended texlive-pictures texlive-plain-generic tipa
xfonts-encodings xfonts-utils
```

Suggested packages:

```
fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
texlive-luatex pandoc-citeproc context wkhtmltopdf librsvg2-bin groff ghc
nodejs php python libjs-mathjax libjs-katex citation-style-language-styles
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
```

```

fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-pstricks
dot2tex prerex texlive-pictures-doc vprerex default-jre-headless tipa-doc
The following NEW packages will be installed:
dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java
libcmark-gfm-extensions0.29.0.gfm libcmark-gfm0.29.0.gfm.3
libcommons-logging-java libcommons-parent-java libfontbox-java libgs9
libgs9-common libidn12 libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java
libptexenc1 libruby3.0 libsynchronet2 libteckit0 libtexlua53 libtexluajit2
libwoff1 libzip-0-13 lmodern pandoc pandoc-data poppler-data
preview-latex-style rake ruby ruby-net-telnet ruby-rubygems ruby-webrick
ruby-xmlrpc ruby3.0 rubygems-integration tlmutils teckit tex-common tex-gyre
texlive texlive-base texlive-binaries texlive-fonts-recommended
texlive-latex-base texlive-latex-extra texlive-latex-recommended
texlive-pictures texlive-plain-generic texlive-xetex tipa xfonts-encodings
xfonts-utils
0 upgraded, 58 newly installed, 0 to remove and 35 not upgraded.
Need to get 202 MB of archives.
After this operation, 728 MB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-droid-fallback all
1:6.0.1r16-1.1build1 [1,805 kB]
Get:2 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-lato all 2.0-2.1
[2,696 kB]
Get:3 http://archive.ubuntu.com/ubuntu jammy/main amd64 poppler-data all
0.4.11-1 [2,171 kB]
Get:4 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-common all 6.17
[33.7 kB]
Get:5 http://archive.ubuntu.com/ubuntu jammy/main amd64 fonts-urw-base35 all
20200910-1 [6,367 kB]
Get:6 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9-common
all 9.55.0~dfsg1-0ubuntu5.12 [753 kB]
Get:7 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libidn12 amd64
1.38-4ubuntu1 [60.0 kB]
Get:8 http://archive.ubuntu.com/ubuntu jammy/main amd64 libijs-0.35 amd64
0.35-15build2 [16.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu jammy/main amd64 libjbig2dec0 amd64
0.19-3build2 [64.7 kB]
Get:10 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgs9 amd64
9.55.0~dfsg1-0ubuntu5.12 [5,031 kB]
Get:11 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 libkpathsea6
amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]
Get:12 http://archive.ubuntu.com/ubuntu jammy/main amd64 libwoff1 amd64
1.0.2-1build4 [45.2 kB]
Get:13 http://archive.ubuntu.com/ubuntu jammy/universe amd64 dvisvgm amd64

```

2.13.1-1 [1,221 kB]
 Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]
 Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]
 Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]
 Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java all 18-1 [4,720 B]
 Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [115 kB]
 Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcmark-gfm-extensions0.29.0.gfm.3 amd64 0.29.0.gfm.3-3 [25.1 kB]
 Get:20 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-java all 43-1 [10.8 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-java all 1.2-2 [60.3 kB]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration all 1.18 [5,336 B]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems all 3.3.5-2ubuntu1.1 [228 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1 [5,100 B]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all 0.1.1-2 [12.6 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsynchronet2 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64 2.5.11+ds1-1 [421 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all

```

1:1.0.5-0ubuntu2 [578 kB]
Get:38 http://archive.ubuntu.com/ubuntu jammy/main amd64 xfonts-utils amd64
1:7.7+6build2 [94.6 kB]
Get:39 http://archive.ubuntu.com/ubuntu jammy/universe amd64 lmodern all
2.004.5-6.1 [9,471 kB]
Get:40 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc-data all
2.9.2.1-3ubuntu2 [81.8 kB]
Get:41 http://archive.ubuntu.com/ubuntu jammy/universe amd64 pandoc amd64
2.9.2.1-3ubuntu2 [20.3 MB]
Get:42 http://archive.ubuntu.com/ubuntu jammy/universe amd64 preview-latex-style
all 12.2-1ubuntu1 [185 kB]
Get:43 http://archive.ubuntu.com/ubuntu jammy/main amd64 t1utils amd64
1.41-4build2 [61.3 kB]
Get:44 http://archive.ubuntu.com/ubuntu jammy/universe amd64 teckit amd64
2.5.11+ds1-1 [699 kB]
Get:45 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tex-gyre all
20180621-3.1 [6,209 kB]
Get:46 http://archive.ubuntu.com/ubuntu jammy-updates/universe amd64 texlive-
binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
Get:47 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-base all
2021.20220204-1 [21.0 MB]
Get:48 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-fonts-
recommended all 2021.20220204-1 [4,972 kB]
Get:49 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-base
all 2021.20220204-1 [1,128 kB]
Get:50 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-
recommended all 2021.20220204-1 [14.4 MB]
Get:51 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive all
2021.20220204-1 [14.3 kB]
Get:52 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libfontbox-java all
1:1.8.16-2 [207 kB]
Get:53 http://archive.ubuntu.com/ubuntu jammy/universe amd64 libpdfbox-java all
1:1.8.16-2 [5,199 kB]
Get:54 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-pictures
all 2021.20220204-1 [8,720 kB]
Get:55 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-latex-extra
all 2021.20220204-1 [13.9 MB]
Get:56 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-plain-
generic all 2021.20220204-1 [27.5 MB]
Get:57 http://archive.ubuntu.com/ubuntu jammy/universe amd64 tipa all 2:1.3-21
[2,967 kB]
Get:58 http://archive.ubuntu.com/ubuntu jammy/universe amd64 texlive-xetex all
2021.20220204-1 [12.4 MB]
Fetched 202 MB in 9s (23.2 MB/s)
Extracting templates from packages: 100%
Preconfiguring packages ...
Selecting previously unselected package fonts-droid-fallback.
(Reading database ... 126374 files and directories currently installed.)

```

```

Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
...
Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Selecting previously unselected package fonts-lato.
Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
Unpacking fonts-lato (2.0-2.1) ...
Selecting previously unselected package poppler-data.
Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
Unpacking poppler-data (0.4.11-1) ...
Selecting previously unselected package tex-common.
Preparing to unpack .../03-tex-common_6.17_all.deb ...
Unpacking tex-common (6.17) ...
Selecting previously unselected package fonts-urw-base35.
Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
Unpacking fonts-urw-base35 (20200910-1) ...
Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.12_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.12_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...

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Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcmark-gfm0.29.0.gfm.3:amd64.
Preparing to unpack .../17-libcmark-gfm0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcmark-gfm-
extensions0.29.0.gfm.3:amd64.
Preparing to unpack .../18-libcmark-gfm-
extensions0.29.0.gfm.3_0.29.0.gfm.3-3_amd64.deb ...
Unpacking libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../19-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../20-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../21-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.
Preparing to unpack .../22-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../23-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../24-ruby-rubygems_3.3.5-2ubuntu1.1_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.1) ...
Selecting previously unselected package ruby.
Preparing to unpack .../25-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../26-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../27-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../28-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../29-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../30-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...

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Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../31-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../32-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../33-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluajit2:amd64.
Preparing to unpack
.../34-libtexluajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../35-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../36-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.
Preparing to unpack .../37-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../38-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package pandoc-data.
Preparing to unpack .../39-pandoc-data_2.9.2.1-3ubuntu2_all.deb ...
Unpacking pandoc-data (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package pandoc.
Preparing to unpack .../40-pandoc_2.9.2.1-3ubuntu2_amd64.deb ...
Unpacking pandoc (2.9.2.1-3ubuntu2) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../41-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package t1utils.
Preparing to unpack .../42-t1utils_1.41-4build2_amd64.deb ...
Unpacking t1utils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../43-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../44-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../45-texlive-

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binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../46-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../47-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../48-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../49-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../50-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../51-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../52-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../53-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../54-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...
Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../55-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../56-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../57-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...

```



```

Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up libcmark-gfm0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up libcmark-gfm-extensions0.29.0.gfm.3:amd64 (0.29.0.gfm.3-3) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up pandoc-data (2.9.2.1-3ubuntu2) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsystex2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.12) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up pandoc (2.9.2.1-3ubuntu2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin
(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps

```

```

tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Collecting py pandoc
  Downloading py pandoc-1.15-py3-none-any.whl.metadata (16 kB)
  Downloading py pandoc-1.15-py3-none-any.whl (21 kB)
Installing collected packages: py pandoc
Successfully installed py pandoc-1.15
Collecting datascience
  Downloading datascience-0.18.0-py3-none-any.whl.metadata (910 bytes)
Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-
packages (from datascience) (0.20.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from datascience) (75.2.0)
Requirement already satisfied: matplotlib>=3.0.0 in
/usr/local/lib/python3.12/dist-packages (from datascience) (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(from datascience) (2.2.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from datascience) (1.16.1)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from datascience) (2.0.2)
Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-
packages (from datascience) (7.34.0)
Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
(from datascience) (5.24.1)
Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages
(from datascience) (0.8.1)
Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (3.1.6)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (2.32.4)
Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-
packages (from folium>=0.9.1->datascience) (2025.4.0)
Requirement already satisfied: contourpy>=1.0.1 in
/usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
(1.3.3)
Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-

```

packages (from matplotlib>=3.0.0->datascience) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (4.59.2)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (1.4.9)
 Requirement already satisfied: packaging>=20.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (25.0)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.0.0->datascience) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (3.2.3)
 Requirement already satisfied: python-dateutil>=2.7 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (2.9.0.post0)
 Collecting jedi>=0.16 (from ipython->datascience)
 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
 Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (4.4.2)
 Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.7.5)
 Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (5.7.1)
 Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
 /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
 Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (2.19.2)
 Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.2.0)
 Requirement already satisfied: matplotlib-inline in
 /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.1.7)
 Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (4.9.0)
 Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
 packages (from pandas->datascience) (2025.2)
 Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
 packages (from pandas->datascience) (2025.2)
 Requirement already satisfied: tenacity>=6.2.0 in
 /usr/local/lib/python3.12/dist-packages (from plotly->datascience) (8.5.0)
 Requirement already satisfied: parso<0.9.0,>=0.8.4 in
 /usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
 (0.8.5)
 Requirement already satisfied: MarkupSafe>=2.0 in
 /usr/local/lib/python3.12/dist-packages (from
 jinja2>=2.9->folium>=0.9.1->datascience) (3.0.2)

```

Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.13)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.3)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.8.3)
Downloading datascience-0.18.0-py3-none-any.whl (725 kB)
725.0/725.0 kB
15.0 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
65.0 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.0 jedi-0.19.2

```

```

[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')

```

Mounted at /content/drive

```

[3]: # Don't change this cell; just run it.
import numpy as np
from datascience import *

```

2 1. Review: The Building Blocks of Python Code

The two building blocks of Python code are *expressions* and *statements*. An **expression** is a piece of code that

- is self-contained, meaning it would make sense to write it on a line by itself, and
- usually evaluates to a value.

Here are two expressions that both evaluate to 3:

```
3
5 - 2
```

One important type of expression is the **call expression**. A call expression begins with the name of a function and is followed by the argument(s) of that function in parentheses. The function returns some value, based on its arguments. Some important mathematical functions are listed below.

Function	Description
<code>abs</code>	Returns the absolute value of its argument
<code>max</code>	Returns the maximum of all its arguments
<code>min</code>	Returns the minimum of all its arguments
<code>pow</code>	Raises its first argument to the power of its second argument
<code>round</code>	Rounds its argument to the nearest integer

Here are two call expressions that both evaluate to 3:

```
abs(2 - 5)
max(round(2.8), min(pow(2, 10), -1 * pow(2, 10)))
```

The expression `2 - 5` and the two call expressions given above are examples of **compound expressions**, meaning that they are actually combinations of several smaller expressions. `2 - 5` combines the expressions `2` and `5` by subtraction. In this case, `2` and `5` are called **subexpressions** because they're expressions that are part of a larger expression.

A **statement** is a whole line of code. Some statements are just expressions. The expressions listed above are examples.

Other statements *make something happen* rather than *having a value*. For example, an **assignment statement** assigns a value to a name.

A good way to think about this is that we're **evaluating the right-hand side** of the equals sign and **assigning it to the left-hand side**. Here are some assignment statements:

```
height = 1.3
the_number_five = abs(-5)
absolute_height_difference = abs(height - 1.688)
```

An important idea in programming is that large, interesting things can be built by combining many simple, uninteresting things. The key to understanding a complicated piece of code is breaking it down into its simple components.

For example, a lot is going on in the last statement above, but it's really just a combination of a few things. This picture describes what's going on.

Question 1.1. In the next cell, assign the name `new_year` to the **larger number** among the following two numbers:

- the **absolute value** of $2^6 - 2^{11} - 2^5 - 7$, and
- $5 \times 13 \times 31 + 9$.

Try to use just one statement (one line of code). Be sure to check your work by executing the test cell afterward.

```
[4]: new_year = max(abs(2**6 - 2**1 - 2**5 - 7), 5*18*31*9)
new_year
```

```
[4]: 25110
```

We've asked you to use one line of code in the question above because it only involves mathematical operations. However, more complicated programming questions will require more steps. It isn't always a good idea to jam these steps into a single line because it can make the code harder to read and harder to debug.

Good programming practice involves splitting up your code into smaller steps and using appropriate names. You'll have plenty of practice in the rest of this course!

3 1.2 Programing Practices: Variable Renaming

As a note, it is important not to use variable names that are the same as an existing function name. Take the `max` function for example.

Currently, it works as expected.

```
[5]: print(max) # just shows us that this is currently a function
max(5, 6, 7) # returns 7, as expected
```

```
<built-in function max>
```

```
[5]: 7
```

If we use `max` as a variable name, we will no longer be able to use the `max` function (thus, don't do this!)

```
[6]: max = 8 # you should not do this!
print(max) # it now prints the number 8, since that was what was stored in it
max(5, 6, 7) # it no longer works!
```

```
8
```

```
-----
TypeError                                Traceback (most recent call last)
/tmp/ipython-input-4066630218.py in <cell line: 0>()
      1 max = 8 # you should not do this!
      2 print(max) # it now prints the number 8, since that was what was stored
    ↪in it
----> 3 max(5, 6, 7) # it no longer works!

TypeError: 'int' object is not callable
```

Run this cell below to revert the max function back to normal for later use:

```
[7]: del max
```

4 2. Importing Code

Most programming involves work that is very similar to work that has been done before. Since writing code is time-consuming, it's good to rely on others' published code when you can. Rather than copy-pasting, Python allows us to **import modules**. A module is a file with Python code that has defined variables and functions. By importing a module, we are able to use its code in our own notebook.

Python includes many useful modules that are just an **import** away. We'll look at the **math** module as a first example. The **math** module is extremely useful in computing mathematical expressions in Python.

Suppose we want to very accurately compute the area of a circle with a radius of 5 meters. For that, we need the constant π , which is roughly 3.14. Conveniently, the **math** module has **pi** defined for us. Run the following cell to import the math module:

```
[10]: import math
      radius = 5
      area_of_circle = radius**2 * math.pi
      area_of_circle
```

```
[10]: 78.53981633974483
```

In the code above, the line **import math** imports the math module. This statement creates a module and then assigns the name **math** to that module. We are now able to access any variables or functions defined within **math** by typing the name of the module followed by a dot, then followed by the name of the variable or function we want.

<module name>.<name>

Question 2.1. The module **math** also provides the name **e** for the base of the natural logarithm, which is roughly 2.71. Compute $e^\pi - \pi$, giving it the name **near_twenty**.

*Remember: You can access **pi** from the **math** module as well!*

```
[11]: near_twenty = math.e**math.pi - math.pi
      near_twenty
```

```
[11]: 19.99909997918947
```

4.1 2.1. Accessing Functions

In the question above, you accessed variables within the **math** module.

Modules also define **functions**. For example, **math** provides the name **floor** for the floor function. Having imported **math** already, we can write **math.floor(7.5)** to compute the floor of 7.5. (Note that the floor function returns the largest integer less than or equal to a given number.)

Question 2.1.1. Compute the floor of pi using `floor` and `pi` from the `math` module. Give the result the name `floor_of_pi`.

```
[12]: floor_of_pi = math.floor(math.pi)
      floor_of_pi
```

```
[12]: 3
```

For your reference, below are some more examples of functions from the `math` module.

Notice how different functions take in different numbers of arguments. Often, the [documentation](#) of the module will provide information on how many arguments are required for each function.

Hint: If you press `shift+tab` while next to the function call, the documentation for that function will appear.

```
[13]: # Calculating logarithms (the logarithm of 8 in base 2).
      # The result is 3 because 2 to the power of 3 is 8.
      math.log(8, 2)
```

```
[13]: 3.0
```

```
[14]: # Calculating square roots.
      math.sqrt(5)
```

```
[14]: 2.23606797749979
```

There are various ways to import and access code from outside sources. The method we used above — `import <module_name>` — imports the entire module and requires that we use `<module_name>.<name>` to access its code.

We can also import a specific constant or function instead of the entire module. Notice that you don't have to use the module name beforehand to reference that particular value. However, you do have to be careful about reassigning the names of the constants or functions to other values!

```
[15]: # Importing just cos and pi from math.
      # We don't have to use `math.` in front of cos or pi
      from math import cos, pi
      print(cos(pi))

      # We do have to use it in front of other functions from math, though
      math.log(pi)
```

```
-1.0
```

```
[15]: 1.1447298858494002
```

Or we can import every function and value from the entire module.

```
[16]: # Lastly, we can import everything from math using the *
# Once again, we don't have to use 'math.' beforehand
from math import *
log(pi)
```

```
[16]: 1.1447298858494002
```

Don't worry too much about which type of import to use. It's often a coding style choice left up to each programmer. In this course, you'll always import the necessary modules when you run the setup cell (like the first code cell in this lab).

Let's move on to practicing some of the table operations you've learned in lecture!

5 3. Table Operations

The table `farmers_markets.csv` contains data on farmers' markets in the United States. Each row represents one such market.

Run the next cell to load the `farmers_markets` table. There will be no output – no output is expected as the cell contains an assignment statement. An assignment statement does not produce any output (it does not yield any value).

```
[19]: # Just run this cell

farmers_markets = Table.read_table('/content/drive/MyDrive/lab03(SSU)/
↳farmers_markets.csv')
farmers_markets
```

```
[19]: FMID      | MarketName                                | street
      | city      | County                                | State      | zip  | x      |
      | y      | Website                                |            |      |      |
      | Twitter                                | Youtube | OtherMedia
      | Organic | Tofu | Bakedgoods | Cheese | Crafts | Flowers | Eggs | Seafood |
Herbs | Vegetables | Honey | Jams | Maple | Meat | Nursery | Nuts | Plants |
Poultry | Prepared | Soap | Trees | Wine | Coffee | Beans | Fruits | Grains |
Juices | Mushrooms | PetFood | WildHarvested | updateTime | Location
      | Credit | WIC | WICcash | SFMNP | SNAP | Season1Date |
Season1Time | Season2Date | Season2Time |
Season3Date | Season3Time | Season4Date | Season4Time
1012063 | Caledonia Farmers Market Association - Danville | nan
      | Danville | Caledonia | Vermont | 05828 | -72.1403 |
44.411 | https://sites.google.com/site/caledoniafarmersmarket/ |
https://www.facebook.com/Danville.VT.Farmers.Market/ | nan
      | nan | nan |
      | N | Y | Y | Y | Y | Y | N | Y | Y
      | Y | Y | Y | Y | N | N | Y | Y | Y
      | Y | N | Y | Y | Y | N | Y | N | Y
      | N | 6/28/2016 12:10:09 PM | nan
```

Y	Y	N	Y	N	06/08/2016 to 10/12/2016	Wed: 9:00 AM-1:00 PM;	nan	nan	nan	
nan	nan	nan								
1011871	Stearns Homestead Farmers' Market				6975 Ridge Road					
Parma	Cuyahoga		Ohio		44130	-81.7286				
41.3751	http://Stearnshomestead.com				nan					
nan			nan	nan						
-	N	Y	N	N	Y	Y	N	Y		
Y	Y	Y	Y	Y	N	N	Y	N		
N	N	N	N	N	N	Y	N	N	N	
Y	N		4/9/2016 8:05:17 PM	nan						
Y	Y	N	Y	Y	06/25/2016 to 10/01/2016	Sat: 9:00 AM-1:00 PM;	nan	nan	nan	
nan	nan	nan								
1011878	100 Mile Market				507 Harrison St					
Kalamazoo	Kalamazoo		Michigan		49007	-85.5749				
42.296	http://www.pfcmarkets.com									
https://www.facebook.com/100MileMarket/?fref=ts					nan					
nan	https://www.instagram.com/100milemarket/				N					
N	Y	Y	N	Y	Y	N	Y	Y		
Y	Y	Y	Y	N	N	N	Y	Y	Y	
N	Y	N	N	Y	Y	N	N	N		
N		4/16/2016 12:37:56 PM	nan							
Y	Y	N	Y	Y	05/04/2016 to 10/12/2016	Wed: 3:00 PM-7:00 PM;	nan	nan	nan	
nan	nan	nan								
1009364	106 S. Main Street Farmers Market				106 S. Main Street					
Six Mile	nan		South Carolina		29682	-82.8187				
34.8042	http://thetownofsixmile.wordpress.com/				nan					
nan			nan	nan						
-	N	N	N	N	N	N	N	N	N	
N	N	N	N	N	N	N	N	N	N	
N	N	N	N	N	N	N	N	N	N	
N	N		2013	nan						
Y	N	N	N	N	nan			nan		
nan	nan	nan	nan	nan	nan	nan	nan			
1010691	10th Steet Community Farmers Market				10th Street and Poplar					
			Lamar	Barton						
Missouri	64759	-94.2746	37.4956	nan						
nan					nan					
nan	http://agrimissouri.com/mo-grown/grodetail.php?type=mo-g	-								
N	Y	N	Y	N	Y	N	Y	Y		
Y	Y	N	Y	N	N	Y	Y	Y	Y	
N	N	N	N	Y	N	N	N	N		
N		10/28/2014 9:49:46 AM	nan							
Y	N	N	N	N	04/02/2014 to 11/30/2014	Wed: 3:00 PM-6:00 PM;Sat: 8:00 AM-1:00 PM;	nan	nan	nan	


```
| N          | 4/3/2014 3:43:31 PM | On a farm from: a barn, a greenhouse,
a tent, a stand, etc | N          | N          | N          | N          | Y          | 05/16/2014 to
10/17/2014 | Fri: 8:00 AM-11:00 AM; | nan        | nan
| nan        | nan        | nan        | nan
1008071 | 14&U Farmers' Market | 1400 U Street NW
| Washington | District of Columbia | District of Columbia | 20009 | -77.0321 |
38.917 | nan
https://www.facebook.com/14UFarmersMarket |
https://twitter.com/14UFarmersMkt | nan | nan
| Y          | N          | Y          | Y          | N          | Y          | Y          | N          | Y
| Y          | Y          | Y          | N          | Y          | N          | Y          | Y          | Y
N          | N          | N          | N          | N          | Y          | Y          | Y          | Y
| N          | N          | 4/5/2014 1:49:04 PM | Other
| Y          | Y          | Y          | Y          | Y          | 05/03/2014 to 11/22/2014 | Sat: 9:00
AM-1:00 PM; | nan        | nan        | nan        |
nan        | nan        | nan
... (8536 rows omitted)
```

Let's examine our table to see what data it contains.

Question 3.1. Use the method `show` to display the first 5 rows of `farmers_markets`.

Note: The terms “method” and “function” are technically not the same thing, but for the purposes of this course, we will use them interchangeably.

Hint: `tbl.show(3)` will show the first 3 rows of the table named `tbl`. Additionally, make sure not to call `.show()` without an argument, as this will crash your kernel!

```
[28]: farmers_markets.show(5)
```

```
<IPython.core.display.HTML object>
```

Notice that some of the values in this table are missing, as denoted by “nan.” This means either that the value is not available (e.g. if we don't know the market's street address) or not applicable (e.g. if the market doesn't have a street address). You'll also notice that the table has a large number of columns in it!

5.0.1 num_columns

The table property `num_columns` returns the number of columns in a table. (A “property” is just a method that doesn't need to be called by adding parentheses.)

Example call: `tbl.num_columns` will return the number of columns in a table called `tbl`

Question 3.2. Use `num_columns` to find the number of columns in our farmers' markets dataset.

Assign the number of columns to `num_farmers_markets_columns`.

```
[26]: num_farmers_markets_columns = farmers_markets.num_columns
print("The table has", num_farmers_markets_columns, "columns in it!")
```

The table has 59 columns in it!

5.0.2 num_rows

Similarly, the property `num_rows` tells you how many rows are in a table.

```
[27]: # Just run this cell

num_farmers_markets_rows = farmers_markets.num_rows
print("The table has", num_farmers_markets_rows, "rows in it!")
```

The table has 8546 rows in it!

5.0.3 select

Most of the columns are about particular products – whether the market sells tofu, pet food, etc. If we’re not interested in that information, it just makes the table difficult to read. This comes up more than you might think, because people who collect and publish data may not know ahead of time what people will want to do with it.

In such situations, we can use the table method `select` to choose only the columns that we want in a particular table. It takes any number of arguments. Each should be the name of a column in the table. It returns a new table with only those columns in it. The columns are in the order *in which they were listed as arguments*.

For example, the value of `farmers_markets.select("MarketName", "State")` is a table with only the name and the state of each farmers’ market in `farmers_markets`.

Question 3.3. Use `select` to create a table with only the name, city, state, latitude (y), and longitude (x) of each market. Call that new table `farmers_markets_locations`.

Hint: Make sure to be exact when using column names with `select`; double-check capitalization!

```
[35]: farmers_markets_locations = farmers_markets.select("MarketName", "city", "State", "y", "x")
farmers_markets_locations
```

```
[35]: MarketName | city | State
      | y | x
      |-----|-----|-----|
      Caledonia Farmers Market Association - Danville | Danville | Vermont
      | 44.411 | -72.1403
      Stearns Homestead Farmers' Market | Parma | Ohio
      | 41.3751 | -81.7286
      100 Mile Market | Kalamazoo | Michigan
      | 42.296 | -85.5749
      106 S. Main Street Farmers Market | Six Mile | South Carolina
      | 34.8042 | -82.8187
      10th Steet Community Farmers Market | Lamar | Missouri
      | 37.4956 | -94.2746
      112st Madison Avenue | New York | New York
      | 40.7939 | -73.9493
      12 South Farmers Market | Nashville | Tennessee
      | 36.1184 | -86.7907
```

```

125th Street Fresh Connect Farmers' Market      | New York    | New York
| 40.809    | -73.9482
12th & Brandywine Urban Farm Market              | Wilmington  | Delaware
| 39.7421   | -75.5345
14&U Farmers' Market                            | Washington  | District of
Columbia | 38.917   | -77.0321
... (8536 rows omitted)

```

5.0.4 drop

`drop` serves the same purpose as `select`, but it takes away the columns that you provide rather than the ones that you don't provide. Like `select`, `drop` returns a new table.

Question 3.4. Suppose you just didn't want the `FMID` and `updateTime` columns in `farmers_markets`. Create a table that's a copy of `farmers_markets` but doesn't include those columns. Call that table `farmers_markets_without_fmids`.

```
[37]: farmers_markets_without_fmids = farmers_markets.drop("FMID", "updateTime")
farmers_markets_without_fmids
```

```
[37]: MarketName      | street
| city      | County      | State      | zip  | x      |
y      | Website      | Facebook
| Twitter      | Youtube | OtherMedia
| Organic | Tofu | Bakedgoods | Cheese | Crafts | Flowers | Eggs | Seafood |
Herbs | Vegetables | Honey | Jams | Maple | Meat | Nursery | Nuts | Plants |
Poultry | Prepared | Soap | Trees | Wine | Coffee | Beans | Fruits | Grains |
Juices | Mushrooms | PetFood | WildHarvested | Location
| Credit | WIC | WICcash | SFMNP | SNAP | Season1Date      |
Season1Time      | Season2Date | Season2Time |
Season3Date | Season3Time | Season4Date | Season4Time
  Caledonia Farmers Market Association - Danville | nan
| Danville | Caledonia      | Vermont      | 05828 | -72.1403 |
44.411 | https://sites.google.com/site/caledoniafarmersmarket/ |
https://www.facebook.com/Danville.VT.Farmers.Market/ | nan
| nan      | nan      | Y
| N      | Y      | Y      | Y      | Y      | Y      | N      | Y      | Y
| Y      | Y      | Y      | Y      | N      | N      | Y      | Y      | Y      | Y
| Y      | N      | Y      | Y      | Y      | N      | Y      | N      | Y
| N      | nan      | Y
| Y      | N      | Y      | N      | 06/08/2016 to 10/12/2016 | Wed: 9:00 AM-1:00
PM;      | nan      | nan      | nan      | nan
| nan      | nan
  Stearns Homestead Farmers' Market      | 6975 Ridge Road
| Parma      | Cuyahoga      | Ohio      | 44130 | -81.7286 |
41.3751 | http://StearnsHomestead.com      | nan
| nan      | nan      | nan
| -      | N      | Y      | N      | N      | Y      | Y      | N      | Y

```

Y	Y	Y	Y	Y	N	N	Y	N	
N	N	N	N	N	N	Y	N	N	N
Y	N		nan						
Y	Y	N	Y	Y	06/25/2016 to 10/01/2016	Sat: 9:00			
AM-1:00 PM;			nan	nan	nan				
nan	nan	nan							
100 Mile Market					507 Harrison St				
Kalamazoo	Kalamazoo		Michigan		49007	-85.5749			
42.296	http://www.pfcmarkets.com								
https://www.facebook.com/100MileMarket/?fref=ts			nan						
nan	https://www.instagram.com/100milemarket/				N				
N	Y	Y	N	Y	Y	N	Y	Y	
Y	Y	Y	Y	N	N	N	Y	Y	Y
N	Y	N	N	Y	Y	N	N	N	
N	nan								Y
Y	N	Y	Y	05/04/2016 to 10/12/2016	Wed: 3:00 PM-7:00				
PM;			nan	nan	nan	nan			
nan	nan								
106 S. Main Street Farmers Market			106 S. Main Street						
Six Mile	nan		South Carolina		29682	-82.8187			
34.8042	http://thetownofsixmile.wordpress.com/		nan						
nan			nan	nan					
-	N	N	N	N	N	N	N	N	N
N	N	N	N	N	N	N	N	N	
N	N	N	N	N	N	N	N	N	N
N	N	nan							
Y	N	N	N	N	nan		nan		
nan	nan	nan	nan	nan	nan	nan	nan		
10th Steet Community Farmers Market			10th Street and Poplar						
Lamar	Barton		Missouri		64759	-94.2746			
37.4956	nan		nan		http://agrimissouri.com/mo-				
grown/grodetail.php?type=mo-g ...	-	N	Y	N	Y	N	Y		
N	Y	N	Y	Y	Y	Y	N	Y	
N	N	Y	Y	Y	Y	N	N	N	N
Y	N	N	N	N	N	nan			
Y	N	N	N	N	04/02/2014 to 11/30/2014	Wed: 3:00			
PM-6:00 PM;Sat: 8:00 AM-1:00 PM;	nan	nan	nan						
nan	nan	nan							
112st Madison Avenue			112th Madison Avenue						
New York	New York		New York		10029	-73.9493			
40.7939	nan		nan		nan				
nan	nan	nan							
-	N	Y	N	Y	Y	N	N	Y	
Y	Y	Y	N	N	N	Y	N	N	
Y	Y	N	N	N	N	N	N	N	N
N	N		Private business parking lot						

N	N	Y	Y	N	July to November	Tue:8:00
am - 5:00 pm;	Sat:8:00 am - 8:00 pm;	nan	nan	nan		
nan	nan	nan				
12 South Farmers Market		3000 Granny White Pike				
Nashville	Davidson	Tennessee	37204	-86.7907		
36.1184	http://www.12southfarmersmarket.com					
12_South_Farmers_Market		@12southfrmsmkt				
nan	@12southfrmsmkt		Y			
N	Y	Y	N	Y	Y	N
Y	Y	Y	Y	N	N	N
N	N	Y	N	Y	N	Y
N	nan					Y
N	N	N	Y	05/05/2015 to 10/27/2015	Tue: 3:30 PM-6:30	
PM;	nan	nan	nan	nan	nan	
nan	nan					
125th Street Fresh Connect Farmers' Market		163 West 125th Street and				
Adam Clayton Powell, Jr. Blvd.	New York	New York	New York			
10027	-73.9482	40.809	http://www.125thStreetFarmersMarket.com			
https://www.facebook.com/125thStreetFarmersMarket						
https://twitter.com/FarmMarket125th	nan	Instagram-->				
125thStreetFarmersMarket	Y	N	Y			
Y	Y	Y	Y	N	Y	Y
Y	Y	N	Y	N	Y	Y
Y	N	Y	N	Y	N	N
Federal/State government building grounds		Y	Y	N		
Y	Y	06/10/2014 to 11/25/2014	Tue: 10:00 AM-7:00 PM;			
nan	nan	nan	nan	nan	nan	
12th & Brandywine Urban Farm Market		12th & Brandywine Streets				
Wilmington	New Castle	Delaware	19801	-75.5345		
39.7421	nan					
https://www.facebook.com/pages/12th-Brandywine-Urban-Far ...	nan					
nan	https://www.facebook.com/delawareurbanfarmcoalition		N			
N	N	N	N	N	N	Y
N	N	N	N	N	N	N
N	N	N	N	Y	N	N
N	On a farm from: a barn, a greenhouse, a tent, a stand, etc		N			
N	N	N	Y	05/16/2014 to 10/17/2014	Fri: 8:00 AM-11:00	
AM;	nan	nan	nan	nan	nan	
nan	nan					
14&U Farmers' Market		1400 U Street NW				
Washington	District of Columbia	District of Columbia	20009	-77.0321		
38.917	nan					
https://www.facebook.com/14UFarmersMarket						
https://twitter.com/14UFarmersMkt	nan	nan				
Y	N	Y	Y	N	Y	N
Y	Y	Y	N	Y	N	Y
N	N	N	N	N	Y	Y

```

| N      | N      | Other
| Y      | Y      | Y      | Y      | 05/03/2014 to 11/22/2014 | Sat: 9:00
AM-1:00 PM; | nan    | nan    | nan    |
nan      | nan    | nan
... (8536 rows omitted)

```

Now, suppose we want to answer some questions about farmers' markets in the US. For example, which market(s) have the largest longitude (given by the `x` column)?

To answer this, we'll sort `farmers_markets_locations` by longitude.

```
[38]: farmers_markets_locations.sort('x')
```

```

[38]: MarketName          | city          | State | y
| x
Trapper Creek Farmers Market | Trapper Creek | Alaska |
53.8748 | -166.54
Kekaha Neighborhood Center (Sunshine Markets) | Kekaha      | Hawaii |
21.9704 | -159.718
Hanapepe Park (Sunshine Markets) | Hanapepe    | Hawaii |
21.9101 | -159.588
Kalaheo Neighborhood Center (Sunshine Markets) | Kalaheo     | Hawaii |
21.9251 | -159.527
Hawaiian Farmers of Hanalei | Hanalei     | Hawaii |
22.2033 | -159.514
Hanalei Saturday Farmers Market | Hanalei     | Hawaii |
22.2042 | -159.492
Kauai Culinary Market | Koloa       | Hawaii |
21.9067 | -159.469
Koloa Ball Park (Knudsen) (Sunshine Markets) | Koloa       | Hawaii |
21.9081 | -159.465
West Kauai Agricultural Association | Poipu       | Hawaii |
21.8815 | -159.435
Kilauea Neighborhood Center (Sunshine Markets) | Kilauea     | Hawaii |
22.2112 | -159.406
... (8536 rows omitted)

```

Oops, that didn't answer our question because we sorted from smallest to largest longitude. To look at the largest longitudes, we'll have to sort in reverse order.

```
[39]: farmers_markets_locations.sort('x', descending=True)
```

```

[39]: MarketName          | city          | State
| y      | x
Christian "Shan" Hendricks Vegetable Market | Saint Croix  | Virgin
Islands | 17.7449 | -64.7043
La Reine Farmers Market | Saint Croix  | Virgin
Islands | 17.7322 | -64.7789

```

```

Anne Heyliger Vegetable Market      | Saint Croix      | Virgin
Islands | 17.7099 | -64.8799
Rothschild Francis Vegetable Market | St. Thomas      | Virgin
Islands | 18.3428 | -64.9326
Feria Agrícola de Luquillo          | Luquillo         | Puerto Rico
| 18.3782 | -65.7207
El Mercado Familiar                 | San Lorenzo      | Puerto Rico
| 18.1871 | -65.9674
El Mercado Familiar                 | Gurabo           | Puerto Rico
| 18.2526 | -65.9786
El Mercado Familiar                 | Patillas         | Puerto Rico
| 18.0069 | -66.0135
El Mercado Familiar                 | Caguas zona urbana | Puerto Rico
| 18.2324 | -66.039
El Maercado Familiar                | Arroyo zona urbana | Puerto Rico
| 17.9686 | -66.0617
... (8536 rows omitted)

```

(The `descending=True` bit is called an *optional argument*. It has a default value of `False`, so when you explicitly tell the function `descending=True`, then the function will sort in descending order.)

5.0.5 sort

Some details about sort:

1. The first argument to `sort` is the name of a column to sort by.
2. If the column has text in it, `sort` will sort alphabetically; if the column has numbers, it will sort numerically - both in ascending order by default.
3. The value of `farmers_markets_locations.sort("x")` is a *copy* of `farmers_markets_locations`; the `farmers_markets_locations` table doesn't get modified. For example, if we called `farmers_markets_locations.sort("x")`, then running `farmers_markets_locations` by itself would still return the unsorted table.
4. Rows always stick together when a table is sorted. It wouldn't make sense to sort just one column and leave the other columns alone. For example, in this case, if we sorted just the `x` column, the farmers' markets would all end up with the wrong longitudes.

Question 3.5. Create a version of `farmers_markets_locations` that's sorted by **latitude** (`y`), with the largest latitudes first. Call it `farmers_markets_locations_by_latitude`.

```

[42]: farmers_markets_locations_by_latitude = farmers_markets_locations.
      ↪sort('y',descending=True)
      farmers_markets_locations_by_latitude

```

```

[42]: MarketName      | city      | State | y      | x
Tanana Valley Farmers Market | Fairbanks | Alaska | 64.8628 | -147.781
Ester Community Market      | Ester     | Alaska | 64.8459 | -148.01
Fairbanks Downtown Market   | Fairbanks | Alaska | 64.8444 | -147.72
Nenana Open Air Market      | Nenana    | Alaska | 64.5566 | -149.096
Highway's End Farmers' Market | Delta Junction | Alaska | 64.0385 | -145.733

```

MountainTraders	Talkeetna	Alaska	62.3231	-150.118
Talkeetna Farmers Market	Talkeetna	Alaska	62.3228	-150.118
Denali Farmers Market	Anchorage	Alaska	62.3163	-150.234
Kenny Lake Harvest II	Valdez	Alaska	62.1079	-145.476
Copper Valley Community Market	Copper Valley	Alaska	62.0879	-145.444

... (8536 rows omitted)

Now let's say we want a table of all farmers' markets in California. Sorting won't help us much here because California is closer to the middle of the dataset.

Instead, we use the table method `where`.

```
[43]: california_farmers_markets = farmers_markets_locations.where('State', are.
      ↪equal_to('California'))
california_farmers_markets
```

```
[43]: MarketName          | city          | State          | y          |
x
Adelanto Stadium Farmers Market | Victorville   | California     | 34.5593    |
-117.405
Alameda Farmers' Market         | Alameda       | California     | 37.7742    |
-122.277
Alisal Certified Farmers' Market | Salinas       | California     | 36.6733    |
-121.634
Altadena Farmers' Market        | Altadena      | California     | 34.2004    |
-118.158
Alum Rock Village Farmers' Market | San Jose      | California     | 37.3678    |
-121.833
Amador Farmers' Market-- Jackson | Jackson       | California     | 38.3488    |
-120.774
Amador Farmers' Market-- Pine Grove | Pine Grove    | California     | 38.3488    |
-120.774
Amador Farmers' Market-- Sutter Creek | Sutter Creek  | California     | 38.3488    |
-120.774
Anderson Happy Valley Farmers Market | Anderson      | California     | 40.4487    |
-122.408
Angels Camp Farmers Market-Fresh Fridays | Angels Camp   | California     | 38.0722    |
-120.543
... (745 rows omitted)
```

Ignore the syntax for the moment. Instead, try to read that line like this:

Assign the name `california_farmers_markets` to a table whose rows are the rows in the `farmers_markets_locations` table **where** the "State" are equal to "California".

5.0.6 where

Now let's dive into the details a bit more. `where` takes 2 arguments:

1. The name of a column. **where** finds rows where that column's values meet some criterion.
2. A predicate that describes the criterion that the column needs to meet.

The predicate in the example above called the function `are.equal_to` with the value we wanted, 'California'. We'll see other predicates soon.

where returns a table that's a copy of the original table, but **with only the rows that meet the given predicate**.

Question 3.6. Use `california_farmers_markets` to create a table called `berkeley_markets` containing farmers' markets in Berkeley, California.

```
[44]: berkeley_markets = california_farmers_markets.where('city', are.
      ↪equal_to('Berkeley'))
      berkeley_markets
```

```
[44]: MarketName          | city      | State      | y          | x
      Downtown Berkeley Farmers' Market | Berkeley | California | 37.8697    | -122.273
      North Berkeley Farmers' Market    | Berkeley | California | 37.8802    | -122.269
      South Berkeley Farmers' Market     | Berkeley | California | 37.8478    | -122.272
```

So far we've only been using **where** with the predicate that requires finding the values in a column to be *exactly* equal to a certain value. However, there are many other predicates. We've listed a few common ones below but if you want a more complete list check out the predicates section on the [python reference sheet](#).

Predicate	Example	Result
<code>are.equal_to</code>	<code>are.equal_to(50)</code>	Find rows with values equal to 50
<code>are.not_equal_to</code>	<code>are.not_equal_to(50)</code>	Find rows with values not equal to 50
<code>are.above</code>	<code>are.above(50)</code>	Find rows with values above (and not equal to) 50
<code>are.above_or_equal_to</code>	<code>are.above_or_equal_to(50)</code>	Find rows with values above 50 or equal to 50
<code>are.below</code>	<code>are.below(50)</code>	Find rows with values below 50
<code>are.between</code>	<code>are.between(2, 10)</code>	Find rows with values above or equal to 2 and below 10
<code>are.between_or_equal_to</code>	<code>are.between_or_equal_to(2, 10)</code>	Find rows with values above or equal to 2 and below or equal to 10

5.1 4. Analyzing a dataset

Now that you're familiar with table operations, let's answer an interesting question about a dataset!

Run the cell below to load the `imdb` table. It contains information about the 250 highest-rated movies on IMDb.

```
[53]: # Just run this cell
```

```
imdb = Table.read_table('/content/drive/MyDrive/lab03(SSU)/imdb.csv')
imdb
```

```
[53]: Votes | Rating | Title | Year | Decade
      88355 | 8.4 | M | 1931 | 1930
      132823 | 8.3 | Singin' in the Rain | 1952 | 1950
      74178 | 8.3 | All About Eve | 1950 | 1950
      635139 | 8.6 | Léon | 1994 | 1990
      145514 | 8.2 | The Elephant Man | 1980 | 1980
      425461 | 8.3 | Full Metal Jacket | 1987 | 1980
      441174 | 8.1 | Gone Girl | 2014 | 2010
      850601 | 8.3 | Batman Begins | 2005 | 2000
      37664 | 8.2 | Judgment at Nuremberg | 1961 | 1960
      46987 | 8 | Relatos salvajes | 2014 | 2010
      ... (240 rows omitted)
```

Often, we want to perform multiple operations - sorting, filtering, or others - in order to turn a table we have into something more useful. You can do these operations one by one, e.g.

```
first_step = original_tbl.where("col1", are.equal_to(12))
second_step = first_step.sort('col2', descending=True)
```

However, since the value of the expression `original_tbl.where("col1", are.equal_to(12))` is itself a table, you can just call a table method on it:

```
original_tbl.where("col1", are.equal_to(12)).sort('col2', descending=True)
```

You should organize your work in the way that makes the most sense to you, using informative names for any intermediate tables you create.

Question 4.1. Create a table of movies released between 2010 and 2015 (inclusive) with ratings above 8. The table should only contain the columns **Title** and **Rating**, **in that order**.

Assign the table to the name `above_eight`.

Hint: Think about the steps you need to take, and try to put them in an order that make sense. Feel free to create intermediate tables for each step, but please make sure you assign your final table the name `above_eight`!

```
[55]: above_eight = imdb.where('Year', are.between_or_equal_to(2010,2015)).
      ↪where('Rating', are.above(8)).select('Title','Rating')
      above_eight
```

```
[55]: Title | Rating
      Gone Girl | 8.1
      Warrior | 8.2
      Intouchables | 8.5
      Shutter Island | 8.1
      12 Years a Slave | 8.1
```

```

Inside Out (2015/I)      | 8.5
Jagten                   | 8.2
Toy Story 3              | 8.3
How to Train Your Dragon | 8.1
Interstellar             | 8.6
... (10 rows omitted)

```

Question 4.2. Use `num_rows` (and arithmetic) to find the *proportion* of movies in the dataset that were released 1900-1999, and the *proportion* of movies in the dataset that were released in the year 2000 or later.

Assign `proportion_in_20th_century` to the proportion of movies in the dataset that were released 1900-1999, and `proportion_in_21st_century` to the proportion of movies in the dataset that were released in the year 2000 or later.

Hint: The *proportion* of movies released in the 1900's is the *number* of movies released in the 1900's, divided by the *total number* of movies.

```

[61]: num_movies_in_dataset = imdb.where('Year', are.between_or_equal_to(1900,3000)).
      ↪ num_rows
      num_in_20th_century = imdb.where('Year', are.between_or_equal_to(1900,1999)).
      ↪ num_rows
      num_in_21st_century = imdb.where('Year', are.between_or_equal_to(2000,2999)).
      ↪ num_rows
      proportion_in_20th_century = num_in_20th_century / num_movies_in_dataset
      proportion_in_21st_century = num_in_21st_century / num_movies_in_dataset
      print("Proportion in 20th century:", proportion_in_20th_century)
      print("Proportion in 21st century:", proportion_in_21st_century)

```

```
Proportion in 20th century: 0.684
```

```
Proportion in 21st century: 0.316
```

5.2 5. Summary

For your reference, here's a table of all the functions and methods we saw in this lab. We'll learn more methods to add to this table in the coming week!

Name	Example	Purpose
<code>sort</code>	<code>tbl.sort("N")</code>	Create a copy of a table sorted by the values in a column
<code>where</code>	<code>tbl.where("N", are.above(2))</code>	Create a copy of a table with only the rows that match some <i>predicate</i>
<code>num_rows</code>	<code>tbl.num_rows</code>	Compute the number of rows in a table
<code>num_columns</code>	<code>tbl.num_columns</code>	Compute the number of columns in a table

Name	Example	Purpose
<code>select</code>	<code>tbl.select("N")</code>	Create a copy of a table with only some of the columns
<code>drop</code>	<code>tbl.drop("N")</code>	Create a copy of a table without some of the columns

Lego wants to congratulate you on finishing Lab 3!

5.3 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit.

Please save before exporting!

```
[ ]: # should change the directory and file name matching to yours
!jupyter nbconvert './DS/lab03(SSU).ipynb' --to pdf
```